

WINDOM, MN, USA

WMO: 727506

Lat: 43.913N

Lon: 95.109W

Elev: 430

StdP: 96.27

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04952

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-22.2	-28.0	0.3	-23.8	-26.0	0.4	-21.7	15.3	-10.5	14.3	-9.2	3.8	320	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.3	23.3	31.0	22.7	28.9	21.8	25.7	29.8	24.5	28.6	23.5	27.3	5.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	20.0	28.0	22.9	18.6	27.0	22.3	18.0	26.4	81.5	30.2	76.8	28.4	72.2	27.5	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.6	10.5	DB	-28.4	35.2	4.6	2.8	-31.6	37.2	-34.3	38.9	-36.9	40.5	-40.2	42.5	(4)
(5)			WB	-28.7	28.0	4.4	2.0	-31.8	29.4	-34.4	30.6	-36.8	31.7	-40.0	33.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-8.7	-7.2	0.2	7.9	14.4	20.1	22.6	20.8	17.0	9.4	1.8	-6.4	(6)	
	DBStd	12.44	7.25	7.23	7.39	5.97	5.13	3.76	3.18	3.20	4.86	5.52	6.65	6.83	(7)	
	HDD10.0	2347	579	483	312	108	17	0	0	0	5	78	256	508	(8)	
	HDD18.3	4245	837	716	562	316	143	24	4	13	83	283	497	767	(9)	
	CDD10.0	1516	0	0	9	44	153	303	391	334	214	59	9	0	(10)	
	CDD18.3	372	0	0	1	2	21	77	137	88	42	6	0	0	(11)	
	CDH23.3	3458	0	0	8	41	241	704	1288	718	392	66	2	0	(12)	
	CDH26.7	1076	0	0	3	9	78	226	435	193	114	17	0	0	(13)	
(14) Wind	WSAvg	4.6	5.2	5.2	5.1	5.4	5.1	4.2	3.4	3.1	3.9	4.5	5.0	4.9	(14)	
(15) Precipitation	PrecAvg	782	19	18	45	82	97	119	101	98	85	57	36	22	(15)	
	PrecMax	1072	55	37	103	191	189	258	160	223	312	161	98	66	(16)	
	PrecMin	576	0	5	0	15	31	61	34	26	14	10	2	4	(17)	
	PrecStd	142	12	10	25	43	41	45	35	53	73	39	29	17	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	14.0	22.5	27.7	32.3	33.8	34.0	32.6	32.3	28.9	21.3	11.3	(19)	
		MCWB	4.4	9.4	14.0	16.2	19.2	22.7	25.6	24.4	21.6	18.5	13.2	8.1	(20)	
	2%	DB	4.0	8.0	17.5	23.6	28.7	31.3	32.4	30.9	29.7	24.8	17.4	7.3	(21)	
		MCWB	1.9	5.7	12.2	13.8	18.1	21.8	24.7	24.2	20.8	16.1	11.6	4.3	(22)	
	5%	DB	2.5	4.2	13.7	21.1	26.2	29.0	31.0	28.8	27.6	22.0	14.0	3.9	(23)	
		MCWB	1.2	2.5	10.0	12.8	17.0	20.7	24.0	22.8	19.7	14.9	9.4	2.1	(24)	
	10%	DB	1.2	2.4	10.9	17.7	22.9	27.5	28.9	27.4	25.9	18.0	11.4	2.4	(25)	
		MCWB	0.2	1.3	8.6	11.2	15.4	19.8	23.1	21.8	18.9	12.7	7.9	1.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	10.7	16.3	18.4	21.9	25.6	27.9	26.6	24.2	21.9	15.2	8.5	(27)	
		MCDB	8.0	13.4	19.5	24.4	28.4	30.1	31.7	30.0	28.7	25.0	19.0	9.7	(28)	
	2%	WB	2.6	5.7	12.9	15.6	19.8	24.0	26.2	25.2	22.9	17.7	12.7	4.4	(29)	
		MCDB	3.8	8.3	16.2	21.2	25.5	28.7	30.4	29.2	27.3	21.4	16.5	6.7	(30)	
	5%	WB	1.4	3.2	10.5	13.8	18.3	22.6	25.1	23.9	21.6	15.5	10.2	2.6	(31)	
		MCDB	2.4	4.5	13.4	19.3	23.8	26.9	29.3	27.4	25.9	20.3	14.1	3.9	(32)	
	10%	WB	0.0	1.2	7.6	12.0	16.9	21.1	24.0	22.8	20.1	13.5	7.8	1.2	(33)	
		MCDB	1.0	2.3	10.4	16.2	21.6	25.9	27.8	26.1	23.9	17.8	10.8	2.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.3	9.4	11.2	11.3	10.7	11.0	10.9	12.4	11.6	10.1	8.6	(35)	
		MCDBR	10.2	10.9	14.8	16.3	15.0	12.9	11.9	12.1	14.1	15.3	14.4	10.7	(36)	
		MCWBR	8.4	8.5	9.4	8.6	7.2	6.1	6.1	6.6	6.8	7.9	9.0	8.5	(37)	
	5% WB	MCDBR	9.4	10.7	13.2	13.7	12.1	10.9	10.6	10.4	11.7	12.6	13.3	9.8	(38)	
		MCWBR	7.8	8.4	9.1	8.3	6.8	6.4	6.4	6.7	6.6	7.6	9.0	8.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.264	0.277	0.307	0.358	0.378	0.397	0.408	0.405	0.362	0.321	0.282	0.264	(40)	
	taud		2.450	2.426	2.453	2.344	2.353	2.357	2.310	2.292	2.390	2.496	2.528	2.470	(41)	
	Ebn at Noon		880	926	937	906	893	874	859	848	863	862	853	845	(42)	
	Edh at Noon		71	87	97	118	121	121	126	123	103	81	66	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.86	2.92	3.82	4.71	5.48	6.09	6.55	5.42	4.31	2.86	1.91	1.42	(44)	
	RadStd		0.20	0.31	0.37	0.45	0.40	0.44	0.25	0.41	0.30	0.37	0.15	0.14	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.89	N/A	N/A	N/A	N/A	
(47) Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page